

Sharp symmetry bounds for real algebraic curves

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Abstract

Let C be an infinite real plane algebraic curve defined by a polynomial irreducible over $\mathbb{C}$, of degree $d \geq 2$, and suppose that C is not a circle. Its rotation group has order at most $\max\{d, 2d - 4\}$, whereas its full Euclidean symmetry group has order at most $2d$. Both bounds are sharp. For $d \geq 5$ we classify the curves attaining the rotation bound: up to orientation-preserving similarity they are $\operatorname{Re}(z^{d-2}(|z|^2 + \alpha)) = 0$, where $|\alpha| = 1$ and $\alpha \notin \mathbb{R}$. Their spherical closures have a dihedral group of $4d - 8$ holomorphic Möbius symmetries and no anti-Möbius symmetry. The parameter is a complete invariant under holomorphic Möbius equivalence, and is determined up to conjugation under full Möbius equivalence. An irreducible quintic with a rotation of order six illustrates why a symmetry of a zero set need not fix its defining polynomial.

1 Statements and context

Throughout, $f \in \mathbb{R}[x, y]$ is irreducible in $\mathbb{C}[x, y]$, $d = \deg f \geq 2$, and $C = \{(x, y) \in \mathbb{R}^2 : f(x, y) = 0\}$ is infinite and is not a circle. The degree is the ordinary total degree, not a degree invariant under Möbius transformations. Write $\operatorname{Sym}(C)$ for the group of Euclidean isometries preserving C , and $\operatorname{Sym}^+(C)$ for its orientation-preserving subgroup. The identity is counted in both groups.

Theorem 1 (Sharp bounds). *The groups $\operatorname{Sym}(C)$ and $\operatorname{Sym}^+(C)$ are finite. If $N = |\operatorname{Sym}^+(C)|$, then*

$$N \leq \max\{d, 2d - 4\}, \quad |\operatorname{Sym}(C)| \leq 2d.$$

Both bounds are attained in every degree $d \geq 2$. If $d \geq 5$ and $N = 2d - 4$, then an orientation-preserving similarity carries C onto

$$C_{m,\alpha} = \{z \in \mathbb{C} : \operatorname{Re}(z^m(|z|^2 + \alpha)) = 0\}, \quad m = d - 2, \quad |\alpha| = 1, \quad \alpha \notin \mathbb{R}. \quad (1)$$

Conversely every curve in (1) with $m \geq 3$ attains that rotation bound.

The coefficient method behind the proof is classical. Lebmeir and Richter-Gebert [1, Section 4], and Lebmeir [2, Theorem 5.4], recover rotational symmetry from the differences of the weights $a - b$ of the nonzero coefficients of $z^a \bar{z}^b$. We use this criterion with irreducibility to obtain degree bounds and their equality cases; we do not propose a new symmetry-detection method. Pach and de Zeeuw [3, Lemma 2.6] use a bound of $4d$ on the number of isometries in a distinct-distance argument. The bound $2d$ above sharpens that auxiliary estimate; no improvement of their distance exponent is claimed.

The distinction between fixing a polynomial and fixing its zero set matters even for an irreducible curve. The quintic

$$f(x, y) = (x^2 + y^2)(x^3 - 3xy^2) - 3x^2y + y^3 = \operatorname{Re}(z^3(|z|^2 + i)) \quad (2)$$

satisfies $f(e^{\pi i/3}z) = -f(z)$ and has exactly six rotations. Its irreducibility follows from Lemma 4 below. Thus it is a counterexample to the degree restriction in [4, Lemma 9], where the direct

symmetry group is asserted to preserve a regular polygon with at most $\deg C$ vertices. This comparison concerns the cited arXiv version; the journal version has not been checked.

The equality cases have more structure than their Euclidean symmetries alone reveal. Let $\widehat{C}_{m,\alpha}$ denote the closure of $C_{m,\alpha}$ in the Riemann sphere. A full Möbius equivalence means either a holomorphic Möbius map or such a map followed by complex conjugation, acting on the ambient sphere.

Theorem 2 (The extremal family). *For every integer $m \geq 2$ and every nonreal α of modulus one, $C_{m,\alpha}$ is an infinite geometrically irreducible curve of degree $m+2$. The full ambient Möbius symmetry group of $\widehat{C}_{m,\alpha}$ consists of*

$$z \mapsto cz \quad (c^{2m} = 1), \quad z \mapsto c/z \quad (c^{2m} = \bar{\alpha}^2). \quad (3)$$

All these maps are holomorphic. The group is dihedral of order $4m$; the Euclidean symmetry group consists of the $2m$ rotations in the first set. For fixed m , the curves $\widehat{C}_{m,\alpha}$ and $\widehat{C}_{m,\beta}$ are holomorphically Möbius equivalent if and only if $\beta = \alpha$, and anti-Möbius equivalent if and only if $\beta = \bar{\alpha}$.

The absence of anti-Möbius symmetries is an ambient statement, not a claim about the absence of a real structure on the complexified curve. Neither theorem concerns arbitrary automorphisms of its normalization.

2 The degree bounds

We first recall the finiteness argument, including the hypotheses needed to pass from a real zero set to a polynomial identity.

Lemma 3. *The group $\text{Sym}(C)$ is finite, its direct subgroup is cyclic, and every $T \in \text{Sym}(C)$ satisfies $f \circ T = \varepsilon(T)f$ with $\varepsilon(T) \in \{1, -1\}$. If T is a reflection then $\varepsilon(T) = 1$.*

Proof. A nonzero translation preserving C would put infinitely many points of C on a line: iterate any point of C along the translation vector. By Bézout and irreducibility C would be that line. A nontrivial glide reflection is excluded by squaring it. Two rotations with distinct centers cannot both preserve C . Indeed, write them as $R_a(z) = \lambda z + (1 - \lambda)a$ and $R_b(z) = \mu z + (1 - \mu)b$, where $\lambda, \mu \neq 1$. Their commutator is a translation, and is nontrivial because $R_a R_b$ and $R_b R_a$ differ by $(1 - \lambda)(1 - \mu)(a - b) \neq 0$. Thus all nonidentity rotations have one center a . The images of any point of $C \setminus \{a\}$ under distinct rotations are distinct points on one circle. Bézout bounds their number by $2d$. If there are no rotations, the direct subgroup is trivial. In either case it is a finite subgroup of the circle group, hence cyclic. The full group has index at most two over this subgroup.

The real locus is infinite, hence Zariski dense in the irreducible complex curve $f = 0$. Consequently $f \circ T$ is a nonzero scalar multiple of f , since these polynomials have the same degree. The scalar is real; iterating the finite-order map T makes it 1 or -1 . If a reflection had scalar -1 , then f would vanish on its entire fixed line. That would give a linear factor, contrary to $d \geq 2$ and irreducibility. $\square$

Identify $z = x + iy$ and set

$$P(X, Y) = f((X + Y)/2, (X - Y)/(2i)) = \sum_{a+b \leq d} p_{ab} X^a Y^b.$$

The polynomial P is irreducible and $p_{ba} = \overline{p_{ab}}$. Translate a rotation center to zero. For a primitive N th root ζ , invariance of C under $z \mapsto \zeta z$ says

$$P(\zeta X, \zeta^{-1} Y) = \varepsilon P(X, Y), \quad p_{ab} \neq 0 \implies \zeta^{a-b} = \varepsilon. \quad (4)$$

Proof of Theorem 1. Finiteness follows from Lemma 3. If $N = 1$ the bounds are immediate, so assume $N > 1$ and apply (4) to a generator. If $\varepsilon = 1$ and $N > d$, every exponent has $a = b$. Thus $P = R(XY)$. Irreducibility forces R to be linear, and the infinite real locus forces C to be a circle, a contradiction. Therefore $\varepsilon = 1$ implies $N \leq d$.

Suppose $N > d$ instead. Then $\varepsilon = -1$, $N = 2m$ is even, and $a - b \equiv m \pmod{2m}$. Since $m > d/2$, the only possible weights are m and $-m$. Hence

$$P = X^m A(XY) + Y^m \bar{A}(XY), \quad \deg A \leq \left\lfloor \frac{d-m}{2} \right\rfloor, \quad (5)$$

where $\bar{A}$ means coefficientwise conjugation. If A were constant, P would be a homogeneous binary form of degree $m \geq 2$, hence reducible. Thus $\deg A \geq 1$, giving $m \leq d-2$ and $N \leq 2d-4$.

If there is a reflection S , every rotation R is a product of the reflections RS and S . Both fix f by Lemma 3, so the generator has $\varepsilon = 1$. Therefore $N \leq d$ and $|\text{Sym}(C)| = 2N \leq 2d$. Without a reflection, $|\text{Sym}(C)| = N \leq \max\{d, 2d-4\} < 2d$.

For sharpness of $2d$, use $\text{Re}(z^d) = 1$ for each $d \geq 2$. Its complexification $X^d + Y^d - 2$ is irreducible by Eisenstein at $Y - \eta$ in $\mathbb{C}[Y][X]$, where $\eta^d = 2$. The real locus is infinite, and its d rotations and d reflections give equality. These examples also attain the rotation bound for $d = 2, 3$. For every $d \geq 4$, take $m = d-2$ and $\alpha = i$ in Theorem 2, whose proof is independent of the present theorem. It gives exactly $2d-4$ rotations.

If $d \geq 5$ and $N = 2d-4 > d$, equation (5) has $m = d-2 \geq 3$ and $A(s) = a + bs$ with $a, b \neq 0$. Here $b \neq 0$ follows from the degree, and $a = 0$ would give a factor XY . If a/b were real, P would factor as $(XY + a/b)(bX^m + \bar{b}Y^m)$. Thus $a/b \notin \mathbb{R}$. Substitute $z = re^{i\phi}w$, choosing $r^2 = |a/b|$ and $e^{im\phi}b$ positive real, and rescale the equation by a nonzero real constant. This gives (1) with $\alpha = (a/b)/|a/b|$. The converse is Theorem 2. $\square$

3 Equality cases on the sphere

Put

$$P_\alpha(X, Y) = X^m(\alpha + XY) + Y^m(\bar{\alpha} + XY), \quad m \geq 2. \quad (6)$$

Its closure V_α in $\mathbb{P}^1 \times \mathbb{P}^1$ has bidegree $(m+1, m+1)$. This compactification, rather than the ordinary projective plane, lets ambient Möbius maps act by regular automorphisms.

Lemma 4. *For $|\alpha| = 1$ and $\alpha \notin \mathbb{R}$, P_α is irreducible. The only singular points of V_α are $(0, 0)$ and (∞, ∞) , both ordinary m -fold points. Its normalization has genus m , and $C_{m, \alpha}$ is infinite.*

Proof. In the chart $X = tY$, remove the exceptional factor Y^m to obtain

$$H(t, Y) = (\alpha t^m + \bar{\alpha}) + t(t^m + 1)Y^2 = 0. \quad (7)$$

The two coefficients in this quadratic in Y are coprime in $\mathbb{C}[t]$: at $t = 0$ the first is nonzero, and at $t^m = -1$ it is $\bar{\alpha} - \alpha \neq 0$. The rational function

$$-\frac{\alpha t^m + \bar{\alpha}}{t(t^m + 1)}$$

has a simple pole at $t = 0$, so it is not a square in $\mathbb{C}(t)$. Gauss's lemma makes H irreducible. The change $X = tY$ is invertible where $Y \neq 0$, and P_α has no factor Y , proving its irreducibility.

On $XY \neq 0$, equation (7) is a local coordinate description. Its Y derivative can vanish only at $t^m = -1$, where $H \neq 0$. Thus there are no singularities there. The tangent cones at $(0, 0)$ and (∞, ∞) are, respectively, $\alpha X^m + \bar{\alpha} Y^m$ and $u^m + v^m$, with $u = 1/X$, $v = 1/Y$. Each has m distinct linear factors. The other two boundary points are $(\infty, 0)$ and $(0, \infty)$. Near the first, the equation is

$$u^{m+1} P_\alpha(1/u, Y) = \alpha u + Y + \bar{\alpha} u^{m+1} Y^m + u^m Y^{m+1},$$

whose linear part is nonzero. The other point is treated by exchanging X, Y and conjugating coefficients. Setting either coordinate to zero or infinity shows that these are all boundary points.

The double cover (7) has $2m + 2$ simple branch points: the m roots of each of its coprime degree- m factors, and $0, \infty$. Riemann–Hurwitz gives $2g - 2 = -4 + (2m + 2)$, so $g = m$. For each $r > 0$ the equation $\operatorname{Re}(e^{im\theta}(r^2 + \alpha)) = 0$ has $2m$ distinct angles modulo 2π , because $r^2 + \alpha \neq 0$. $\square$

Proof of Theorem 2. The degree and real-locus assertions follow from (6) and Lemma 4. A holomorphic map M acts on the complexification by $(X, Y) \mapsto (M(X), \overline{M}(Y))$; an anti-Möbius map $M \circ \operatorname{conj}$ acts by $(X, Y) \mapsto (M(Y), \overline{M}(X))$. Here $\overline{M}$ conjugates the coefficients of M . The infinite real locus is Zariski dense, so an equivalence of the spherical real loci carries V_α onto V_β . It must preserve the unordered pair of singular points. Consequently the ambient map has one of the four forms

$$cz, \quad c/z, \quad c\bar{z}, \quad c/\bar{z} \quad (c \in \mathbb{C}^\times).$$

Write $t = |c|^2$. Substitute each form into P_β , clearing the denominator by $X^{m+1}Y^{m+1}$ when needed. Proportionality to P_α forces the ratio of the coefficients of X^m and $X^{m+1}Y$ to equal α . In the four cases these ratios are

$$\frac{\beta}{t}, \quad \frac{t}{\bar{\beta}}, \quad \frac{\bar{\beta}}{t}, \quad \frac{t}{\beta}, \quad \text{respectively.} \quad (8)$$

Since $|\alpha| = |\beta| = 1$, necessarily $t = 1$. The holomorphic cases therefore require $\alpha = \beta$; the antiholomorphic cases require $\alpha = \bar{\beta}$. These conditions are sufficient: use the identity or conjugation. In particular a self-equivalence cannot be antiholomorphic.

For $\beta = \alpha$, comparison of the remaining coefficients gives $c^{2m} = 1$ for cz , and $c^{2m} = \bar{\alpha}^2$ for c/z . Substitution verifies both identities, proving (3). Choose $\zeta = e^{\pi i/m}$ and $c_0^m = \bar{\alpha}$. The maps $r(z) = \zeta z$ and $s(z) = c_0/z$ satisfy $r^{2m} = s^2 = 1$ and $srs = r^{-1}$; their $4m$ distinct maps are exactly (3). Among them only the rotations are Euclidean isometries, since c/z does not fix infinity. $\square$

Remark 5 (Scope of the classification). The restriction $d \geq 5$ in the equality clause of Theorem 1 is necessary: in degree four, $\operatorname{Re}(z^4) = 1$ also attains the rotation bound and is not in the $m = 2$ family, since its normalization has genus three rather than two. For fixed $m \geq 2$, the parameters $\alpha = e^{i\theta}$, $0 < \theta < \pi$, give pairwise fully Möbius-inequivalent examples. If α is algebraic, all the maps in (3) have algebraic coefficients. These are explicit consequences of the classification, not separate finiteness or descent theorems.

References

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